

NEBULAOPS v22.3 FINAL

Software & Tools Technical Specification

Runtime toolchain, platform services, local build requirements and validation workflow.

53

services

8

volumes

5

core tools

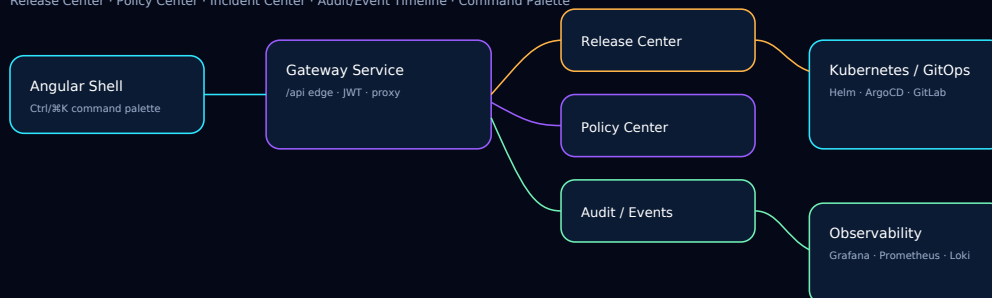
7

new port entries

Professional release closure package - prepared for technical review, handover and local validation.

NebulaOps v22.3 · Operational Intelligence & Governance Layer

Release Center · Policy Center · Incident Center · Audit/Event Timeline · Command Palette



Reference figure - NebulaOps v22.3 platform topology.

1. Software and tools scope

The v22.3 final package preserves the existing toolchain and adds governance/release/audit/notification capabilities. Tooling remains local-first and Docker Compose based, with Kubernetes/Helm/ArgoCD/GitLab assets retained for GitOps projection.

Tool/service	Port	Role
keycloak	8180	OIDC/JWKS identity provider
mongodb	27017	Document persistence
redis	6379	Runtime cache
rabbitmq	5672	Event broker
grafana	3000	Dashboards
prometheus	-	Metrics
loki	3100	Logs
tempo	3200, 4317	Traces
otel-collector	4318	Telemetry pipeline
gitlab	8929, 2222	Optional SCM/CI runtime
gateway-service	8080	API edge
release-orchestrator-service	8098	Release lifecycle
policy-governance-service	8100	Governance policies
audit-service	8101	Audit events

2. Build/runtime requirements

Requirement	Purpose	Command
Java 21	Spring Boot services	java -version
Maven	Backend compilation	mvn -f backend/pom.xml clean package -DskipTests
Node.js + npm	Angular shell/remotes	npm install && npm run build
Docker Compose	Runtime orchestration	docker compose build && docker compose up -d
WSL/Linux shell	Operational scripts	./scripts/wsl/health.sh

3. Validation caveat

Angular builds for the shell and new v22.3 remotes were executed successfully. Maven and Docker were not available in the generation container and must be executed on the target local WSL/Docker environment. The final package includes scripts and runbooks for that validation.

```
mvn -f backend/pom.xml clean package -DskipTests
docker compose -f docker-compose.yml config
docker compose build
docker compose up -d
./scripts/wsl/health.sh
./scripts/wsl/smoke-v22.3.sh
```